MAJOR PROJECT · INDIAN CRIMINAL LAW · LEGAL AI

Pramana

प्रमाण *proof; a means of valid knowledge*

Legal research for Indian criminal law that checks every section it cites, across the switch from the IPC to the BNS. It runs entirely on a laptop.

Your name · Roll number · Department · Guide

On 1 July 2024 India renumbered its criminal law.

INDIAN PENAL CODE, 1860 offences before 1 July 2024	SECTION	BHARATIYA NYAYA SANHITA, 2023 offences from 1 July 2024
Punishment for murder	302	Uttering words with deliberate intent to wound religious feelings
Punishment for murder by a life-convict	303	Theft
Cheating and dishonestly inducing delivery of property	420	*No such section. The BNS ends at 358.*

A section number means nothing without its code, and *the offence date decides the code.*

Language models cite law with confidence, including law that does not exist.

THE MODEL PRAMANA RUNS · QWEN3.5:4B

Q What does BNS Section 303 cover?

A A public-panic offence: making or spreading a bomb threat.

× Wrong. BNS 303 is Theft.

Asked on this machine, 25 Sep 2026. The answer is paraphrased.

20 *of* **68**

citations named sections that do not exist in the BNS when the model answered from memory, mostly IPC numbers such as 420, 378 to 383 and 465 to 468.

AND THE FACTS ARE CONFIDENTIAL

A lawyer types a client’s facts into this form. They should never leave the machine, so there is no cloud model and no API key. Everything runs locally.

WHAT PRAMANA DOES

Facts and a date go in. A checked research memo comes out.

CASE FACTS

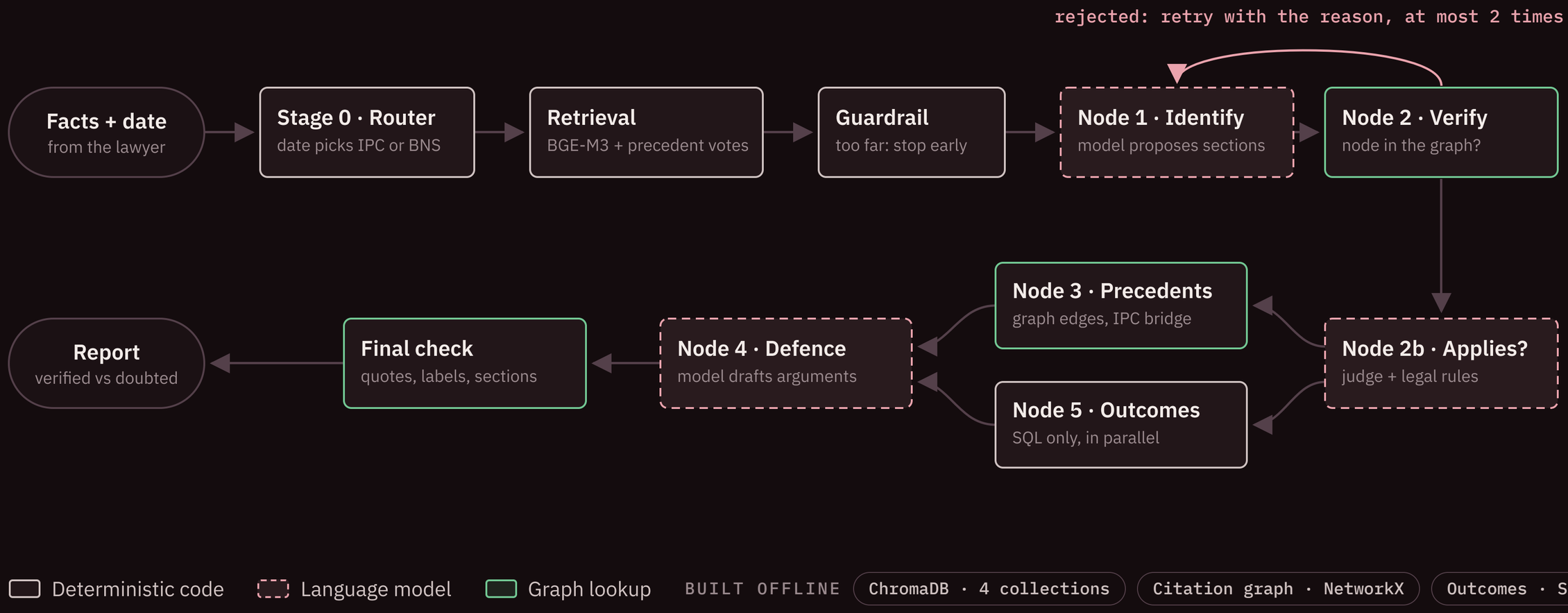
The complainant parked his motorcycle outside his house at night. In the morning it was gone. CCTV shows the accused pushing it away down the lane and riding off. The accused had no permission to take it and did not return it.

OFFENCE DATE 15 Aug 2024

about a minute, offline

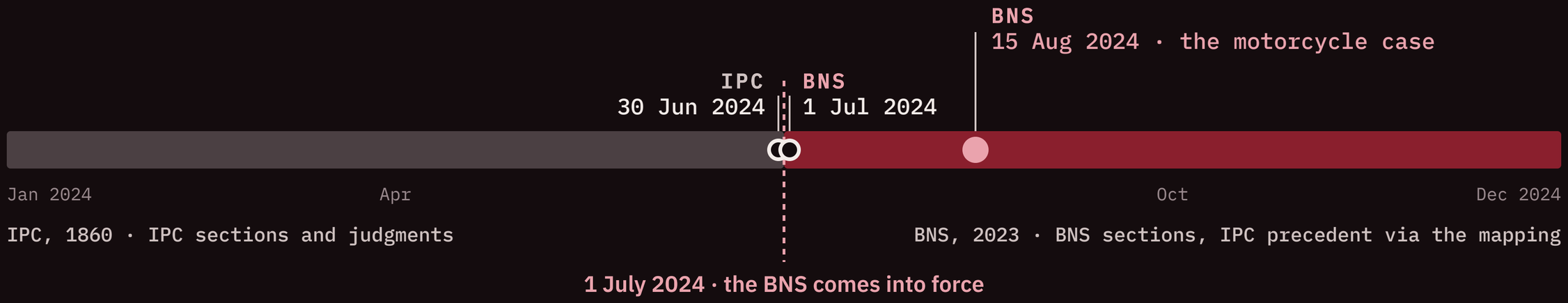
CODE	BNS, because the offence is on or after 1 July 2024
CHARGE	BNS 303 Theft ✓ in the graph
PRECEDENTS	3 judgments, found through its IPC predecessor, IPC 379
DEFENCE	1 counter-argument, its quote checked against the facts
OUTCOMES	How 158 appeals under IPC 379 were decided

The model proposes. The graph decides.



STAGE 0 · THE OFFENCE-DATE ROUTER

The offence date picks the code. No model is involved.

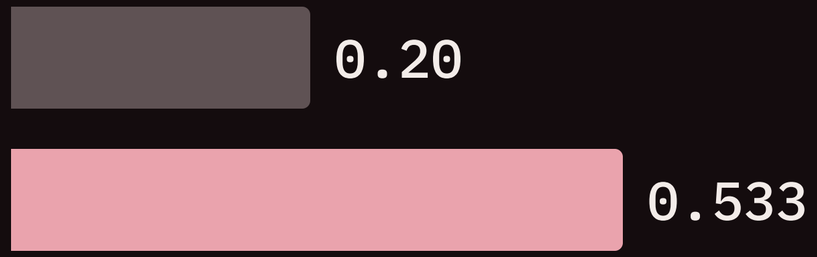



```python
# src/pipeline/router.py
def route(offence_date):
    d = parse_date(offence_date)  # strict ISO date
    return "ipc" if d < transition_date() else "bns"
```

- Runs before anything else. Every later lookup uses only this code.
- An ambiguous date is never guessed. A guessed date is a guessed code.
- A course of conduct that crosses the date is analysed under both codes.

We measured the embedders. The legal-domain model came second.

RECALL@3 ON 15 HAND-WRITTEN TEST CASES · HIGHER IS BETTER



BGE-M3 also led on recall@5 (0.60 against 0.33) and MRR (0.295 against 0.223). It is the embedder for all 87,036 judgment passages and 933 sections.

WHY THE DOMAIN MODEL LOST: HUBNESS

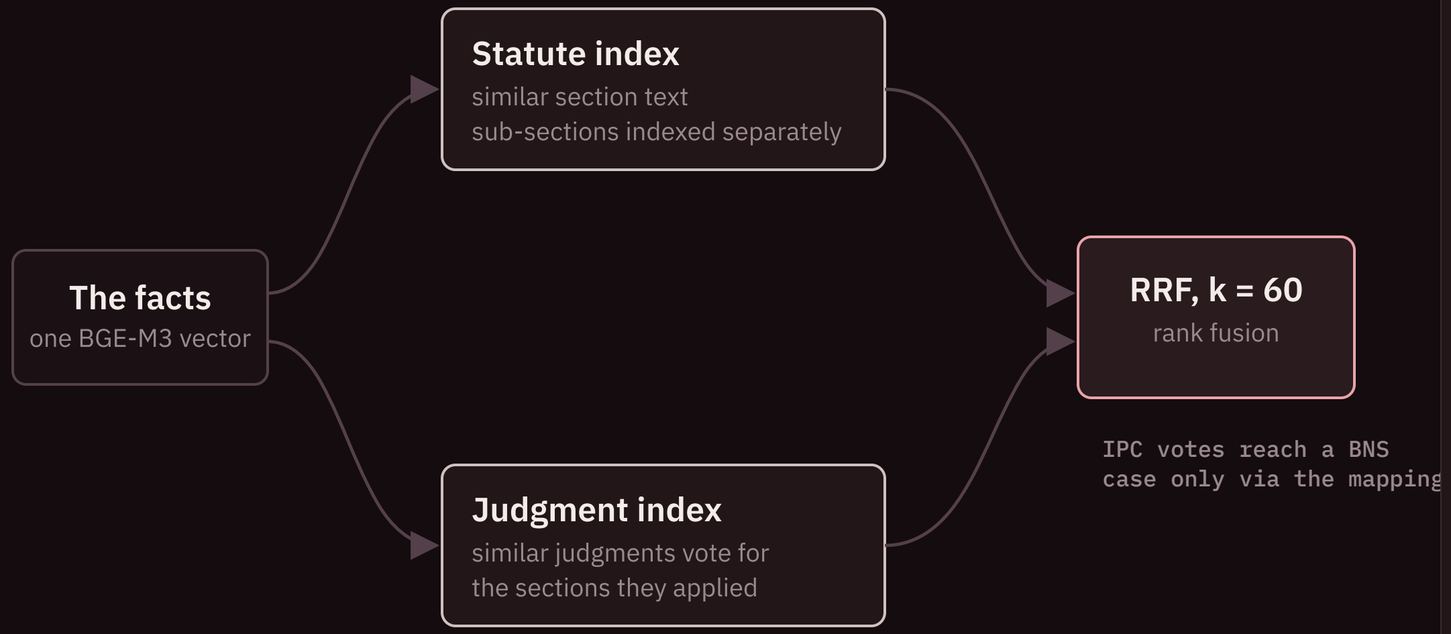
For a motorcycle theft, InLegal-Sbert's top three were:

1. **IPC 443** Lurking house-trespass
2. **IPC 278** Making atmosphere noxious to health
3. **IPC 86** Offence by one who is intoxicated

Theft, **IPC 378 / 379**, first appeared at rank **311**. The same few sections topped almost every query.

RETRIEVAL · TWO SIGNALS, FUSED

Statute text finds what a section says. Precedents show what courts charged.



CANDIDATES FOR THE MOTORCYCLE CASE · BNS

1	303	Theft
2	317	Stolen property
3	324	Mischief
4	238	Causing disappearance of evidence
5	331	House-trespass or house-breaking
6	309	Robbery

Top 6 of 14 shown to Node 1. Real output of the saved sample.

NODE 2 · THE CITATION GRAPH

A section is real only if it is a node of the applicable code.

933

section nodes: all 575 of the IPC and all 358 of the BNS

2,699

judgments, from the Supreme Court and High Courts

16,493

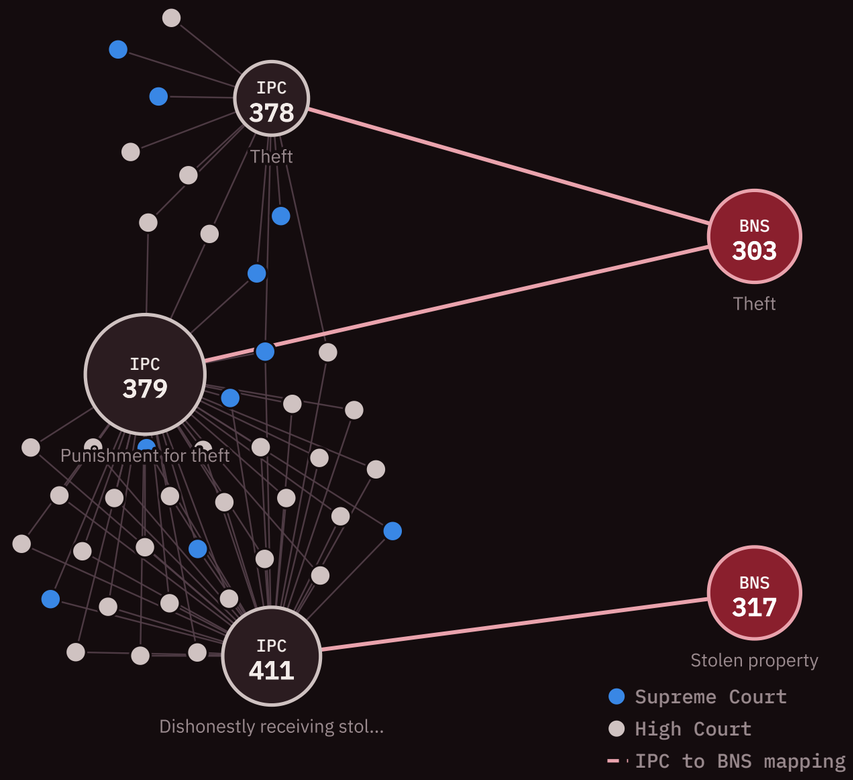
judgment-to-section citation edges

539

reviewed IPC-to-BNS bridges

✓ 20 / 20

Phase 3 gate: 10 real and 10 invented section numbers, including BNS 420 and 498A, real IPC numbers the BNS does not have.



A real piece of the graph: 41 judgments citing IPC 378, 379 and 411. The BNS sections connect only through the mapping.

VERIFICATION · ONE SECTION AT A TIME

How a section earns its place in the report.

RETRIEVED candidate list	PROPOSED by the model	IN THE GRAPH node of the BNS?	APPLIES judge + legal rules
BNS 317 Stolen property	~~BNS 421~~ from memory ✕		BNS 303 Theft ✓

Step 1. The model proposes 303. Under stress it also cites 421 from memory.

Step 2. 303 is a BNS node. 421 is not: the BNS has sections 1 to 358.

Step 3. The judge checks 303's ingredients against the facts. Supported.

NODE 1 AND NODE 2 · THE RETRY LOOP

A rejected citation goes back to the model, with the reason.

ROUND 0 — `model: 421`

✕ BNS 421 is not a section of the ingested BNS (sections 1 to 358).

→

ROUND 1 — `model: 378`

✕ Not a BNS section. IPC 378 (Theft) is an IPC section; under the verified mapping its BNS counterpart is **BNS 303**.

→

ROUND 2 — `model: 303`

✓ **BNS 303 · Theft.** A node of the BNS. Verified.

Phase 4 gate · BNS theft with retrieval withheld · the same path in 3 of 3 runs

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20 / 68

citations rejected when the model cited from memory

6 / 6

affected cases ended on a verified section within 2 retries

0

rejected citations leaked into a report

NODE 2B · APPLICABILITY

A real section can still be the wrong charge.

THE JUDGE, A LANGUAGE MODEL

It reads each verified section's own text against the facts, twice. A section is marked `supported` only if both passes agree. Otherwise it is shown as `doubted`, with the missing ingredient named. Nothing is hidden.

EXAMPLE FROM A CHEATING CASE, CONTEXT WITHHELD

BNS 66, 67, 68 are sexual offences. All three exist, so all three passed the graph. That is why a separate applicability check exists.

DEV SET, BEFORE AND AFTER THE RULES

44% → 58% of supported charges were right

3.5 → 2.6 sections per case

DETERMINISTIC LEGAL RULES, EACH WITH AN OFF SWITCH FOR TESTING

Lesser offence included
417 shown inside 420, 324 inside 326

Definition charged under its punishment
IPC 378 Theft is charged as 379

Injury grade picks the hurt section
weapon, grievous injury, intent

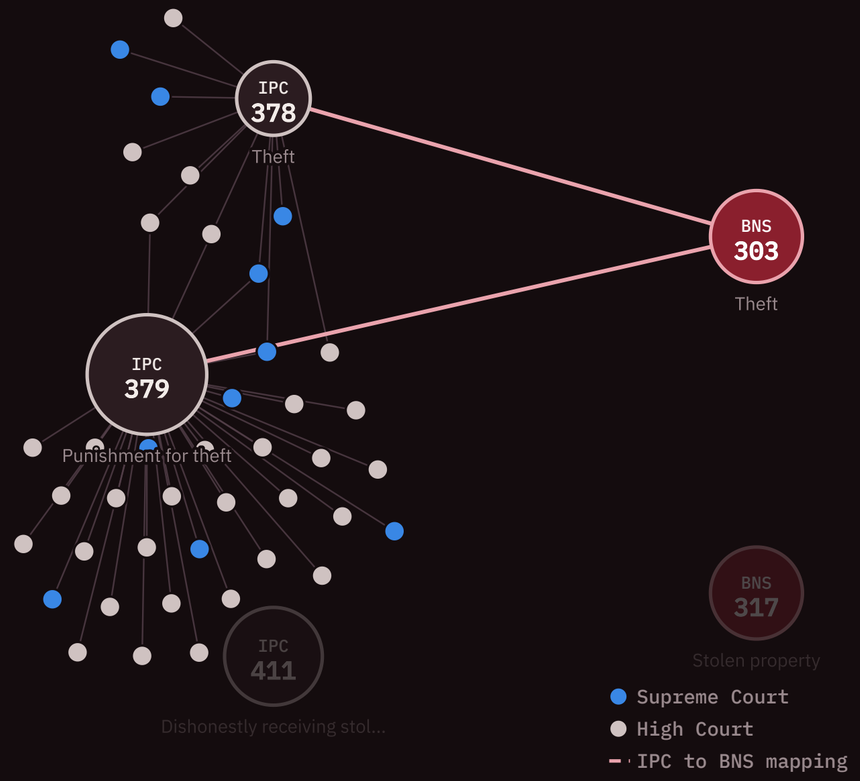
Common intention from the headcount
several accused: IPC 34 / BNS 3(5)

Self-negated proposals withdrawn
"the facts do not disclose..."

Attempt needs no result
attempt to murder is not doubted for "no death"

NODE 3 · PRECEDENTS

No judgment in the corpus applies BNS 303 yet. The mapping bridges the gap.



BNS 303	0	judgments cite it directly
IPC 378	11	via the mapping
IPC 379	158	via the mapping

TOP 3 BY SIMILARITY TO THE FACTS

P1	Telangana High Court	0.686	Allowed
P2	Himachal Pradesh High Court	0.676	Allowed
P3	Delhi High Court	0.658	Dismissed

NODE 4 AND THE FINAL CHECK

Counter-arguments are checked before anyone reads them.

COUNTER-ARGUMENT AGAINST BNS 303 · DRAFTED BY THE MODEL

The prosecution has failed to prove the essential ingredient of dishonest intention *(animus furandi)*. The facts merely state that the accused moved the motorcycle without permission...

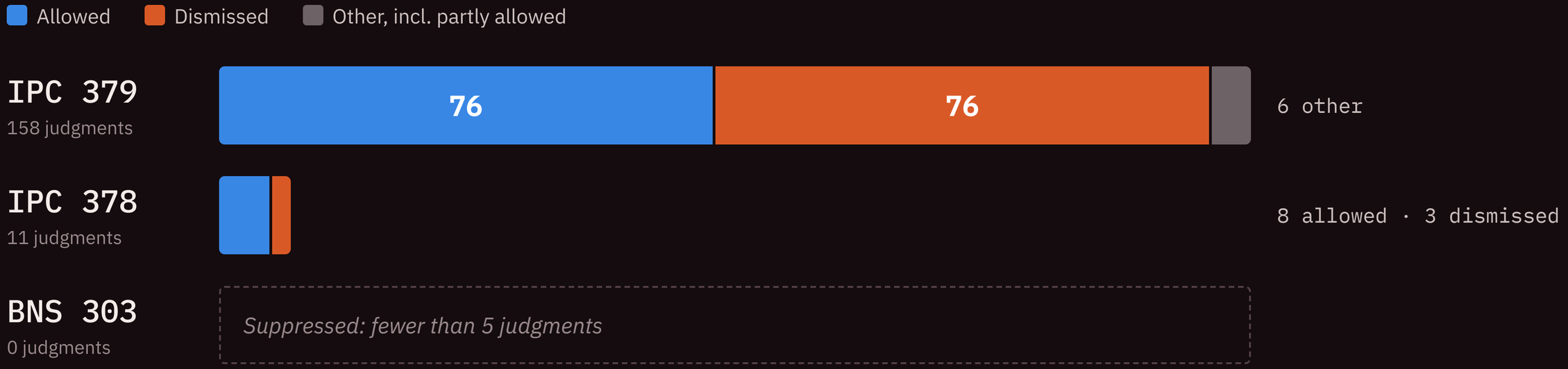
FACT QUOTED

“The accused had no permission to take it and did not return it.”

- **It targets a charge in the report**
  BNS 303 is a verified charge
- **The quote is verbatim**
  found word for word in the lawyer’s facts
- **Precedents cited were supplied**
  only labels P1 to P3 may appear; here none are used
- **A failure is shown, never hidden**
  the argument is marked unverified, with the failed check named

NODE 5 · OUTCOME ANALYTICS

Outcome figures are counted from judgments, never generated.



Allowed means the appellant or petitioner succeeded. This is how appeals were disposed of, **not a conviction rate**.

Pure SQL over a table built offline. No model runs at this step.

THE PRODUCT · THE INTERFACE

A live view of the pipeline, built for lawyers.

▶ VIDEO

Live demo: the motorcycle case, from typing the facts to the report

drop presentation/media/demo.mp4

Live pipeline view
each stage lights up with what it really produced

Verification ledger
every section: retrieved, proposed, in the graph, applies

Citation-graph view
charges, precedents and arguments, linked

Lawyer tools
bare act, IPC-BNS word diff, punishment and bail card, PDF upload, memo

Evaluation dashboard
the measured numbers, inside the product

Verified and doubted are never confused.

SCREENSHOT

Live pipeline view mid-run

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SCREENSHOT

Report: verified charges and doubted ones

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SCREENSHOT

Citation-graph view

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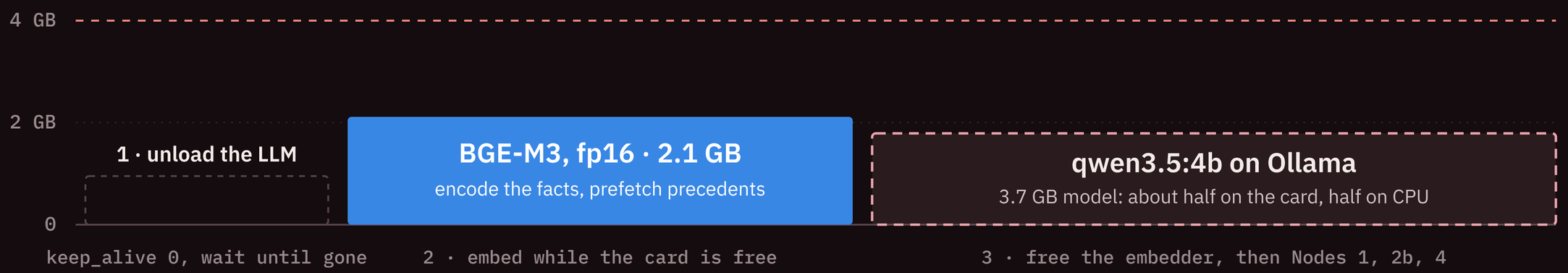
SCREENSHOT

Evaluation dashboard

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ENGINEERING · ONE LAPTOP, ZERO COST

Two models take turns on a 4 GB graphics card.



36.3s → 18.4s

before generation, after moving the query encode from CPU to the freed GPU

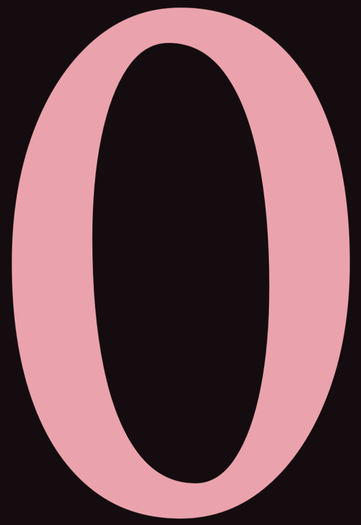
₹0

recurring cost. No API key, no account, works offline

413

automated tests across ingestion, retrieval, graph, pipeline and API

EVALUATION · THE GUARANTEE



invented or wrong-code sections reached a report, in every evaluation we ran.

20 / 20
citation gate: real vs invented section numbers

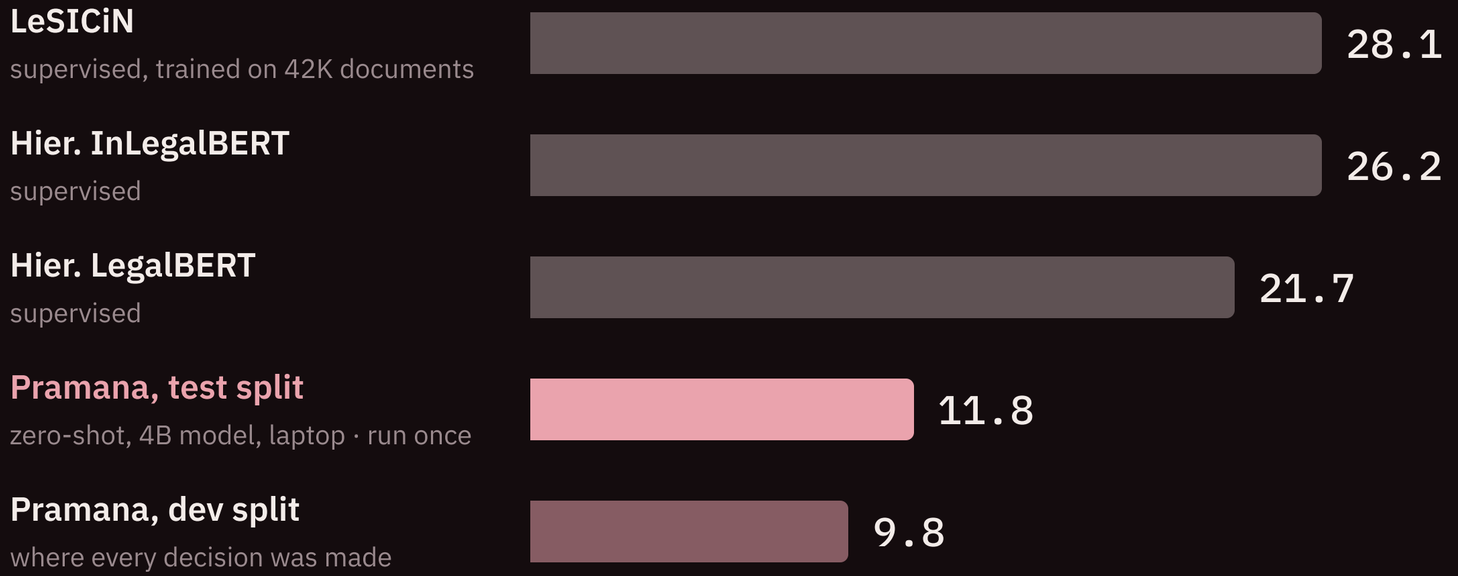
30 / 30
routing gate: 10 cases, 3 runs each, right code every time

0 / 94
proposals rejected on 30 real court orders, so with retrieval the model did not invent

EVALUATION · THE IL-TUR BENCHMARK (ACL 2024)

Choosing the right sections is the weak point. We report it as measured.

LEGAL STATUTE IDENTIFICATION · MACRO-F1 OVER 100 LABELS



100 documents per split, fixed seed. Test 95% CI over labels with support: 13.9 (10.9 to 17.2). Micro P / R / F1: 40 / 21 / 27.

STRONG ON GRAVE, COMMON OFFENCES

302	Murder	0.78
376	Rape	0.63
34	Common intention	0.44

WEAK ON MINOR COMPANION OFFENCES

323	Simple hurt	0.14
341	Wrongful restraint	0.00
504	Intentional insult	0.00

Only 37% of the right sections reached the candidate list.

WHAT DID NOT WORK, AND WHAT IS LEFT

Measured, then removed.

NEGATIVE RESULTS

Legal-domain embedder
recall@3 0.20 vs 0.533; hubness

Cross-encoder reranker
5 / 15 alone, 8 / 15 fused, vs 11 / 15 without it

Asking the model twice, two orderings
precision fell from 35% to 26%

Companion-offence rule
failed rules set before its run; not merged

Fine-tuning
never tried by design: a fine-tuned Indian legal model scored below its base model on AIBE

LIMITATIONS

- Over-charging: real, plausible extra sections (34, 307, 149)
- Minor companion offences rarely retrieved
- The IPC-BNS mapping was reviewed against gazette text, not yet by a practising lawyer
- Counter-argument quality has no metric yet
- Prior-case retrieval is not benchmarked

NEXT

- Retrieval coverage for companion offences
- A lawyer’s review of the mapping and the rules
- Ingest new BNS judgments as they arrive

LIVE DEMO

Let's run it.

THE CASE WE TEST AGAINST

The complainant parked his motorcycle outside his house at night. In the morning it was gone. CCTV shows the accused pushing it away down the lane and riding off. The accused had no permission to take it and did not return it.

OFFENCE DATE 15 August 2024

WATCH FOR router picks BNS | 303 ranked first | verified in the graph | precedents via IPC 379

Thank you.

Pramana: every section it shows exists in the code the offence date selects, and the system says when it doubts one.

RAG, not fine-tuning

BGE-M3 · ChromaDB · NetworkX · LangGraph

qwen3.5:4b on Ollama

FastAPI

Questions welcome.